

AKHILESH KUMAR MISHRA

SOFTWARE DEVELOPMENT ENGINEER | MICROSERVICES · DISTRIBUTED SYSTEMS · FULL STACK

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EXPERIENCE

Software Development Engineer

CSC e-Governance Services India Ltd. (MeitY, Government of India)

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Sustained **500,000+ daily transactions** at a **99% success rate** across AEPS, MATM and UPI, by making the settlement path idempotent with request keys, backoff retries and automated reconciliation.
- Cut p95 read/write latency **45%** across **MySQL**, **PostgreSQL** and **MongoDB**, by sharding and partitioning the hot tables and tuning the slowest queries against their plans.
- Contained a bad release to one service instead of the whole platform for **2 million+ users**, by splitting monolithic backends into **Spring Boot**, **FastAPI** and **Node.js** microservices behind a versioned gateway.
- Lifted message throughput **35%**, by moving inter-service work onto **Apache Kafka**, **Celery** and **Redis** with dead-letter handling for poison messages.
- Blocked **99.9%** of unauthorized access attempts, by adding **JWT** rotation, **RBAC**, device binding and OTP rate limiting across every payment entry point.
- Replaced a manual release process across all **3** payment platforms, by containerising every service with **Docker**, shipping through **GitHub Actions** and putting each one on **Prometheus** and **Grafana** dashboards.
- Improved team onboarding efficiency **40%**, by hiring and onboarding **4 engineers** and writing the runbooks and service documentation they start from.

Software Engineer (Contract)

Independent Client Engagement

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Engineered a workforce telemetry platform for **1,000+ field agents** end to end — **React.js** front end, **FastAPI** backend, AWS deployment.
- Held **99.9% uptime** while processing **10,000+ concurrent** GPS pings, through a **FastAPI**, **Celery** and **Redis** ingestion pipeline.
- Brought dashboard load time down **30%**, by building the live map in **React.js** with **Ant Design** over a WebSocket feed instead of polling.

SELECTED PROJECTS

DigiPay Developer Platform

Monorepo platform pairing a FastAPI service — versioned routers, correlation-ID tracing, auth and rate-limit middleware — with published SDK packages and embeddable React widgets.

github.com/GitHub-akhilesh/Django_apis_digipay

STACK: FastAPI / React / TypeScript / Docker / npm workspaces

Polyglot Microservices Task Platform

Django, Flask and FastAPI services orchestrated under Docker Compose with Kubernetes manifests, fronted by a React web client and a React Native mobile app. github.com/GitHub-akhilesh/To-do-list-app

STACK: Django / Flask / FastAPI / React / React Native / Kubernetes

Transaction Orchestration & Ledger Platform

Microservice platform orchestrating AEPS, MATM and UPI settlement with a transactional wallet ledger, idempotent retries, automated reconciliation and Kafka event fan-out.

STACK: Spring Boot / FastAPI / Kafka / PostgreSQL / Redis

LabX Design System

Industrial data and visual automation project documented end to end — business case, HLD, LLD, data-flow and database design — and published as a live site. github.com/GitHub-akhilesh/labx-design

STACK: Python / System Design / Technical Documentation / GitHub Pages

SUMMARY

Distributed systems for national e-governance and FinTech — 2 million+ users and 500,000+ transactions a day across Spring Boot, FastAPI and Node.js services. Cut p95 latency 45% across MySQL, PostgreSQL and MongoDB, lifted message throughput 35% on Kafka, and put every service behind Prometheus and Grafana. Hired and onboarded 4 engineers and moved the team onto Docker-based CI/CD.

EDUCATION

Bachelor of Technology (B.Tech.)

Kanpur Institute of Technology

2019 – 2023 | Kanpur, UP

SKILLS

LANGUAGES

Java Python JavaScript (ES6+) TypeScript SQL
C++ HTML5 CSS3

BACKEND & ARCHITECTURE

Spring Boot Django FastAPI Node.js Express.js
React.js Next.js microservices REST APIs
idempotency rate limiting WebSockets

DATA & MESSAGING

MySQL PostgreSQL MongoDB Redis Apache Kafka
Celery Hibernate/JPA sharding partitioning
query tuning

TESTING, DEVOPS & OBSERVABILITY

pytest JUnit Jest Docker Kubernetes
GitHub Actions CI/CD Linux Jira Prometheus
Grafana

CERTIFICATIONS

Data Structures & Algorithms Certification
Database Systems & SQL Certification
Advanced MERN Stack Developer Certification
IIT Kanpur Cyber Security Certification

ACHIEVEMENTS

- First Class Honors, B.Tech. — top academic rank in cohort.
- HackerRank 4-Star Gold badge in SQL.